

CODE SPORT



In this month's column, we discuss the solutions to some of the questions we discussed in last month's column.

As you know, it is our practice to feature computer science interview questions in the column regularly, for our student readers to prepare for interviews. A few of our student readers have requested me to provide the answers to the questions as well. Given that this column is meant to kindle the reader's curiosity and seek out solutions instead of providing canned answers to questions, I generally do not provide solutions to the questions featured here. So, instead of providing the complete solutions, as requested, it may help the readers to discuss the high level solutions in more detail, while they work out the concrete answers. Hence, in this month's column, we will discuss some of the topics that the featured questions are based on.

One of the questions from last month's column was on the difference between virtual machines and containers. Virtual machines are intended to facilitate server virtualisation, wherein the physical hardware resources are virtualised by a special piece of software known as the virtual machine monitor or hypervisor. This controls the virtual machines that run on top of it. Each virtual machine has its own copy of its operating system. Typically, a VMM or hypervisor runs on top of the host operating system (though it is possible to have a bare metal hypervisor which runs on top of the firmware itself). Each VM runs its own operating system, which is known as the guest operating system. These guest operating systems can be different from each other. On the other hand, containers run on top of a single host operating system, which is why they are sometimes referred to as supporting operating system virtualisation, whereas virtual machines are referred to as supporting server virtualisation. Typically, multiple containers sit on top of a single

host operating system, which itself runs on top of a physical server. All the containers share the same host operating system kernel and the libraries.

Now, let us consider the case in which we have a bug in an operating system. If this operating system version was running as a guest OS in a VM environment, the impact of the OS bug would be confined only to that VM running the guest OS. On the other hand, assume that this OS was the host OS on top of which all containers were hosted. All the containers would be impacted by the same bug, since there is no OS isolation in a container environment. In that case, what then is the advantage of containers? The very fact that containers share the same OS components makes them very lightweight and easy to deploy. The container density is much higher than the VM density, given the same physical configuration. A detailed discussion of the security issues of containers can be found in the blog post <https://opensource.com/business/14/7/docker-security-selinux>.

Question (3) from last month's column was about information retrieval. Readers were asked why we need to use inverse document frequency as a measure for relevant documents retrieval, given a query. As we know, TF-IDF is primarily used as the measure for information retrieval. TF stands for term frequency. This is a measure of how frequently a term occurs in a single document. If T1 is a query term and it occurs with a high frequency in document D1, it seems obvious that D1 may be a relevant document when a user is searching for query T1. However, let us consider that our document collection is a set of documents associated with cricket, and our query is 'wisden player'. Now, it is possible that many documents contain the query term 'player'